Aspectual Counting — Axioms, Theorems, Dynamics, and Random Aspects

Arithmetic Metaphysics — canvas packet

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0) Core data and notation (ACA)

Let E be a set of **tokens**. An **aspect** α on E is an equivalence relation $\approx_\alpha$. Write $U_\alpha := E/\approx_\alpha$ and $\pi_\alpha : E \to U_\alpha$ for the quotient map.

For finite $X \subseteq E$, the **aspectual count** is

$$a(X, \alpha) := |\pi_\alpha[X]| \in \mathbb{N}.$$

An α**-automorphism** is a function $t : E \to E$ such that $x \approx_\alpha y \iff t(x) \approx_\alpha t(y)$ and the induced $\bar{t} : U_\alpha \to U_\alpha$, $\bar{t}([x]) = [t(x)]$, is bijective. The set of these forms a group $\mathrm{Aut}_\alpha(E)$.

Refinement. For aspects α, β, define $\beta \preceq \alpha$ iff $\approx_\beta \subseteq \approx_\alpha$ (β is finer than α). Then there is a canonical surjection $\tau_{\beta\to\alpha} : U_\beta \twoheadrightarrow U_\alpha$ with $\pi_\alpha = \tau_{\beta\to\alpha} \circ \pi_\beta$.

Meet/Join of aspects. $\alpha \wedge \beta$ has relation $\approx_\alpha \cap \approx_\beta$ (finest common refinement). $\alpha \vee \beta$ is the equivalence closure of $\approx_\alpha \cup \approx_\beta$ (coarsest common coarsening).

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1) Axioms and basic theorems

Axiom A (Counting-by-quotient)

For all finite $X \subseteq E$, $a(X, \alpha) = |\pi_\alpha[X]|$.

Axiom B (Aspect symmetry)

If $t \in \mathrm{Aut}_\alpha(E)$, then $a(t[X], \alpha) = a(X, \alpha)$ for all finite $X \subseteq E$.

Lemma 1 (Aspectual symmetry invariance). Follows immediately from Axiom B / bijectivity of $\bar{t}$:

$$\pi_\alpha[t[X]] = \bar{t}[\pi_\alpha[X]] \Rightarrow |\pi_\alpha[t[X]]| = |\pi_\alpha[X]|.$$

Axiom C (Refinement)

If $\beta \preceq \alpha$, then $\tau_{\beta\to\alpha}$ exists as above.

Lemma 2 (Refinement monotonicity). If $\beta \preceq \alpha$, then for finite X,

$$a(X,\beta) \geq a(X,\alpha).$$

Proof. $\tau_{\beta\to\alpha}(\pi_\beta[X]) = \pi_\alpha[X]$ and surjections do not increase cardinality.

Union identity (overlap penalty). With $O_\alpha(X,Y) := \{u \in U_\alpha : u \cap X \neq \varnothing \,\&\, u \cap Y \neq \varnothing\}$,

$$a(X \cup Y,\alpha) = a(X,\alpha) + a(Y,\alpha) - |O_\alpha(X,Y)|.$$

Bounds via meet/join. For all finite X:

$$a(X,\alpha \wedge \beta) \geq \max\{a(X,\alpha), a(X,\beta)\}, \qquad a(X,\alpha \vee \beta) \leq \min\{a(X,\alpha), a(X,\beta)\}.$$

Extremes. Identity aspect id: $x \approx_{\mathsf{id}} y \iff x = y \Rightarrow a(X,\mathsf{id}) = |X|$. Chaotic aspect $\top$: all tokens equivalent $\Rightarrow a(X,\top) = \mathbf{1}_{X \neq \varnothing}$.

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2) Measure-valued aspectual counting (infinite sets)

Let (E,Σ) be measurable. Equip U_α with the quotient σ-algebra $\Sigma_\alpha = \{S \subseteq U_\alpha : \pi_\alpha^{-1}(S) \in \Sigma\}$.

2.1 Pure unit-count on the quotient

Counting measure ν_α on (U_α,Σ_α):

$$a_\infty(X,\alpha) := \nu_\alpha(\pi_\alpha[X]) \in \mathbb{N} \cup \{\infty\}.$$

Invariance and refinement monotonicity carry over verbatim.

2.2 Hit measure relative to a base measure

Given σ-finite μ on E, push forward: $\mu_\alpha = (\pi_\alpha)_*\mu$. Define

a^{\mathrm{hit}}_\mu(X,\alpha):=\mu_\alpha(\pi_\alpha[X])=\mu\big(\text{KATEX51-saturation of }X\big).

If t is α-automorphic and μ-preserving, invariance holds. If $\beta \preceq \alpha$ and quotient measures are consistent ($\mu_\alpha = (\tau_{\beta\to\alpha})_*\mu_\beta$), then refinement monotonicity holds.

2.3 Fiberwise valuations via disintegration

When μ disintegrates along π_α, $\mu = \int \mu_u \, d\mu_\alpha(u)$. For monotone $\varphi\colon[0,\infty) \to [0,\infty)$ with $\varphi(0) = 0$, define

$$A_{\alpha,\varphi}(X) := \int_{U_\alpha} \varphi\big(\mu_u(X)\big)\, d\eta_\alpha(u),$$

with $\eta_\alpha = \mu_\alpha$ (mass-weighted) or ν_α (unit-weighted). Special cases recover hit-counts and total mass.

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3) Dynamic aspects (time/context index)

Let $(C, \le)$ index contexts. A **dynamic aspect family** is $c \mapsto \alpha_c$. Two regimes:

(Refining over time) If $c_1 \le c_2 \Rightarrow \alpha_{c_2} \preceq \alpha_{c_1}$, then for any X, $a_\infty(X, \alpha_c)$ is nondecreasing in c.

(Coarsening over time) If $c_1 \le c_2 \Rightarrow \alpha_{c_1} \preceq \alpha_{c_2}$, then $a_\infty(X, \alpha_c)$ is nonincreasing.

For finite E, if only finitely many merges/splits occur on compact intervals, $c \mapsto a(X, \alpha_c)$ is piecewise-constant (càdlàg). Dynamic symmetries $t_c \in \mathrm{Aut}_{\alpha_c}(E)$ give pointwise invariance.

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4) Categorical wrapper (initiality of the quotient)

Let Quot(E) have objects quotients $\pi : E \to U$ and morphisms $h : U_\alpha \to U_\beta$ with $h \circ \pi_\alpha = \pi_\beta$ (so β coarsens α).

Define Inv$_\alpha$: objects (V, f) with $x \approx_\alpha y \Rightarrow f(x) = f(y)$; morphisms $m : (V, f) \to (W, g)$ satisfying $g = m \circ f$.

Theorem (Initiality). (U_α, π_α) is initial in Inv$_\alpha$. For any (V, f) there is a unique $\bar{f} : U_\alpha \to V$ with $f = \bar{f} \circ \pi_\alpha$.

Meets/joins as limits/colimits. $\alpha \wedge \beta$ is the pullback partition; $\alpha \vee \beta$ the pushout (equivalence closure of the union).

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5) Random aspects (distributions over partitions)

A **random aspect** is a random equivalence relation (partition) Π on E. For finite E and $X \subseteq E$, let

$$K_X := |\{\text{blocks of } \Pi \text{ that intersect } X\}|.$$

5.1 Paintbox (Kingman) representation

If assignments are i.i.d. from a probability vector $(p_i)_{i \ge 1}$ plus **dust** mass $p_0 \in [0, 1]$ (each dust draw forms a fresh singleton), then for $|X| = m$:

$$\mathbb{E}[K_X] = m\, p_0 \; + \; \sum_{i \ge 1} \big(1 - (1 - p_i)^m\big).$$

For the **purely atomic** case ($p_0 = 0$),

$$\mathrm{Var}(K_X) = \sum_{i \geq 1} \big(1 - (1-p_i)^m\big)(1-p_i)^m + 2\sum_{i<j} \Big((1-p_i-p_j)^m - (1-p_i)^m(1-p_j)^m\Big).$$

(For $p_0 > 0$, the same covariance structure applies among atoms; dust contributes an additional binomial component whose moments can be added via the law of total variance.)

5.2 Ewens/CRP ($\theta > 0$)

For the Ewens sampling formula (Chinese Restaurant Process with concentration θ), the number of blocks among m draws has the independent Bernoulli representation $K_m = \sum_{i=1}^m B_i$ with $\mathbb{P}(B_i = 1) = \theta/(\theta + i - 1)$. Hence

$$\mathbb{E}[K_m] = \sum_{i=1}^m \frac{\theta}{\theta + i - 1}, \qquad \mathrm{Var}(K_m) = \sum_{i=1}^m \frac{\theta}{\theta+i-1}\Big(1 - \frac{\theta}{\theta+i-1}\Big),$$

with asymptotics $\mathbb{E}[K_m] \sim \theta \log m$, $\mathrm{Var}(K_m) \sim \theta \log m$.

(These formulas give the expected/typical aspectual count when the aspect itself is random and exchangeable.)

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6) Tiny gallery of examples

(a) Cells on $\mathbb{R}$

Aspect α_{cell}: $x \approx y \iff \lfloor x \rfloor = \lfloor y \rfloor$. For $X = [0, 2.3)$:

$$a_\infty(X, \alpha_{\text{cell}}) = 3, \quad a^{\text{hit}}_{\text{Leb}}(X, \alpha_{\text{cell}}) = 3, \quad \int \mu_u(X)\, d\mu_\alpha(u) = \mathrm{Leb}(X) = 2.3.$$

(b) Graph-connected components

Let E be vertices of a graph G. Aspect α_{conn}: vertices equivalent iff connected in G. For $X \subseteq V$, a(X, \alpha_\text{conn}}) counts the number of connected components touched by X. Refinement monotonicity matches the effect of splitting/merging components.

(c) Time-varying partitions (households)

Tokens are persons; α_t groups co-residents at time t. Mergers/splits of households make $t \mapsto a(X, \alpha_t)$ piecewise-constant; refining vs coarsening regimes model gaining vs losing information.

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7) References (selected)

- S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Springer, 1998.

- V. I. Bogachev, *Measure Theory*, Vol. I–II, Springer, 2007. (quotients, disintegration)
- G. B. Folland, *Real Analysis: Modern Techniques and Their Applications*, 2nd ed., Wiley, 1999. (quotient σ-algebras, pushforward)
- J. F. C. Kingman, "The Representation of Partition Structures," *J. London Math. Soc.* (2) 18 (1978), 374–380. (paintbox)
- J. Pitman, *Combinatorial Stochastic Processes*, Springer LNM 1875, 2006. (Ewens, exchangeable partitions)
- O. Kallenberg, *Foundations of Modern Probability*, 2nd ed., Springer, 2002. (regular conditional probabilities, disintegration)
- L. Lovász, *Large Networks and Graph Limits*, AMS, 2012. (graph components as partitions)

Notes

- All proofs are short and rely on quotients/partitions and functorial pushforwards. Random-aspect formulas are standard in occupancy/paintbox representations. For dust-bearing models, expectations decompose neatly (each dust draw creates a singleton block); variances can be combined via the law of total variance.